

Business Plan for Decision Tree Classification Assignment

BankWise Customer Analytics: Credit Card Offer Acceptance Prediction

Educational synthetic business case and dataset guide

1. Business Overview

BankWise is a mid-sized retail banking company that wants to improve how it promotes new credit card products to existing customers. The bank has several credit card offers, including cashback, travel rewards, low-interest cards, and balance-transfer offers. Instead of contacting all customers with the same message, BankWise wants to use customer data to identify customers who are more likely to accept a specific offer.

The business goal is to support more efficient customer acquisition, reduce wasted marketing effort, and improve the relevance of offers sent to customers. The company also wants students and analysts to interpret model results in a business-friendly way, not only calculate model metrics.

2. Business Problem

BankWise currently spends time and money sending product offers to customers who may not be interested. Low acceptance rates can increase marketing costs, reduce customer satisfaction, and make branch or call-centre teams less efficient. The company needs a decision-support model that can predict whether a customer is likely to accept a credit card offer.

The model is not intended to make final decisions automatically. Instead, it should help marketing and customer relationship teams prioritize outreach and understand which customer characteristics are associated with higher offer acceptance.

3. Machine Learning Decision

Machine learning can help BankWise answer the following business question: "Which customers are most likely to accept a credit card offer?"

For this assignment, students should train a Decision Tree classification model. The model predicts the target variable `Accepted_Offer` using customer profile, account behaviour, previous campaign response, and offer information.

4. Dataset and Prediction Target

Item	Description
Dataset name	BankWise Credit Card Offer Acceptance Dataset
Business domain	Retail banking, customer analytics, product promotion
Machine learning task	Binary classification
Target variable	Accepted_Offer
Positive class	Yes: the customer accepted the credit card offer
Negative class	No: the customer did not accept the offer
Model for assignment	Decision Tree Classifier
Business decision supported	Prioritizing which customers should receive outreach for a credit card offer

5. Input Features

The dataset includes a mix of numerical and categorical variables. Students should inspect the dataset, check missing values and duplicates, handle categorical variables, and then train the Decision Tree model.

Feature	Business Meaning
Age	Customer age in years.
Income_Level	Estimated customer income segment.
Account_Tenure_Years	Length of the banking relationship.
Average_Monthly_Balance	Average account balance.
Monthly_Transactions	Typical number of monthly transactions.
Digital_Banking_Usage	Customer use of online or mobile banking.
Credit_Score_Band	Grouped credit score category.
Existing_Loan	Whether the customer already has a loan.
Has_Savings_Account	Whether the customer has a savings account.
Previous_Campaign_Response	Whether the customer responded to a previous campaign.
Service_Complaints_6M	Number of service complaints in the last six months.
Branch_Visits_6M	Number of branch visits in the last six months.
Marketing_Channel	Channel used for customer outreach.
Offer_Type	The type of credit card offer presented.

6. Why the Prediction Is Useful

- Marketing teams can focus on customers who are more likely to accept the offer.
- The bank can reduce unnecessary calls, messages, and emails to customers who are unlikely to respond.
- Customer relationship managers can tailor conversations based on the features that appear important in the model.
- The company can compare different outreach channels and offer types to improve future campaigns.
- The model can support business interpretation by showing which features are most important in the Decision Tree.

7. Business Interpretation Guidance for Students

After training the Decision Tree model, students should connect the technical results to the business plan. For example, accuracy alone may not be enough. If the bank has a limited marketing budget, precision may be very important because the bank wants the customers predicted as likely to accept to actually accept. If the bank wants to avoid missing many potential customers, recall may also be important. F1-score can be useful when students want a balanced view of precision and recall.

Model Result	Business Interpretation
True Positive	The model predicts that a customer will accept the offer, and the customer actually accepts it.
True Negative	The model predicts that a customer will not accept the offer, and the customer does not accept it.
False Positive	The model predicts that a customer will accept the offer, but the customer does not accept it. This may waste marketing resources.
False Negative	The model predicts that a customer will not accept the offer, but the customer would have accepted it. This means a missed sales opportunity.
Feature importance	Important features can help explain what factors are associated with offer acceptance and guide future campaigns.

8. Possible Model Limitations and Bias

This is a synthetic educational dataset, so it does not represent real BankWise customers. In real business settings, customer data can contain historical bias, incomplete information, or patterns that reflect unequal access to financial products. For example, variables related to income level or credit score band may reflect broader social and economic inequalities. Students should avoid treating the model as a perfectly objective decision-maker.

Another limitation is that customer preferences can change over time. A customer who rejected one offer may accept a different offer later. Therefore, model results should be combined with human judgment, updated data, and responsible business policies.

9. Responsible AI and Human Judgment

Human judgment should still be used because the model only learns patterns from the dataset. It cannot fully understand customer needs, fairness concerns, recent life changes, or the broader business context. BankWise should use the model as a decision-support tool, not as an automatic system for deciding who deserves financial products.

Students should include a short Responsible AI reflection in their notebook or README. They should discuss limitations, possible bias, privacy issues, and why model predictions should be reviewed by people before being used in real customer decisions.

10. Suggested README Summary

Students can use the following short structure in their GitHub README:

- Business problem: Predict whether a customer will accept a credit card offer.
- Dataset: BankWise synthetic customer analytics dataset.
- Model: Decision Tree classification model.
- Target variable: Accepted_Offer.
- Main metrics: confusion matrix, accuracy, precision, recall, F1-score, training accuracy, and testing accuracy.
- Business insight: Use model results and feature importance to improve campaign targeting.
- Limitation: Synthetic data and potential bias in customer-related variables.
- Responsible AI note: Human review is needed before using predictions for business decisions.